Virtualization And Cloud Computing



Group Members:	Humairaa Ganey – 260038113
	Azola Magxunyana – 261008331
	Abongile Mditshwa – 221342028
	Siyolise Moyikwa – 260805610
	Sinomusa Mroqoza – 260144002
	Inathi Pondo – 251030784

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Lecturer:	Mr. Shaheem Sadien
Class Group:	1G

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STEP 1: Installing Oracle VirtualBox

We downloaded and installed Oracle VirtualBox on the host machine. Before installation, I confirmed that virtualization support was enabled in the BIOS and in Task Manager. This ensures the Virtual Machine can run efficiently.

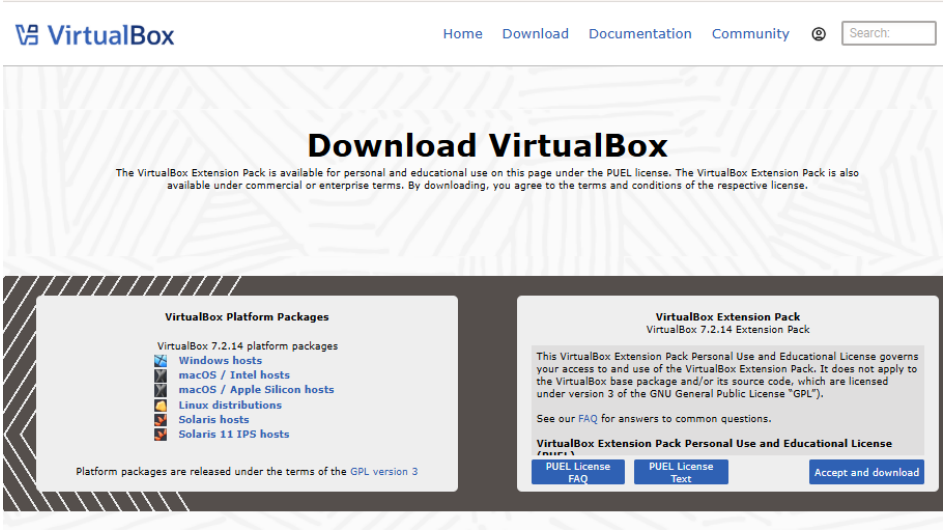


Figure 1: VirtualBox download page

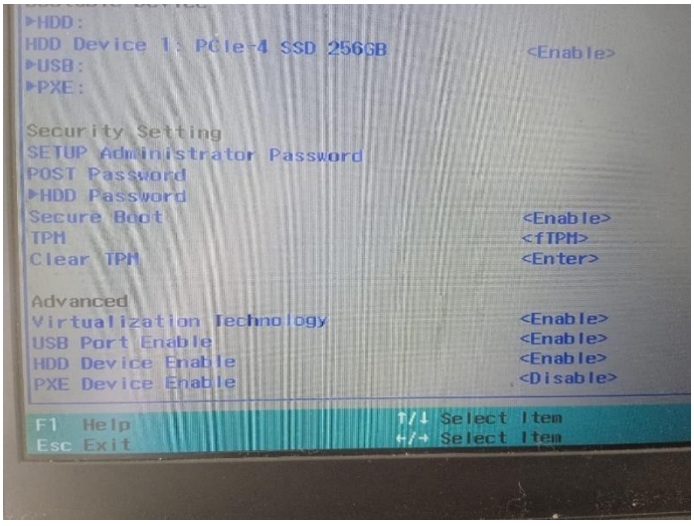


Figure 4: BIOS Virtualization enabled

Explanation:

VirtualBox is the software that allows us to create and run Virtual Machines. BIOS virtualization must be enabled for 64-bit Virtual Machines to work.

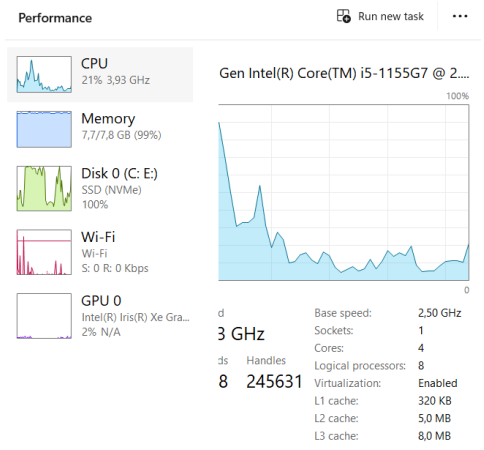


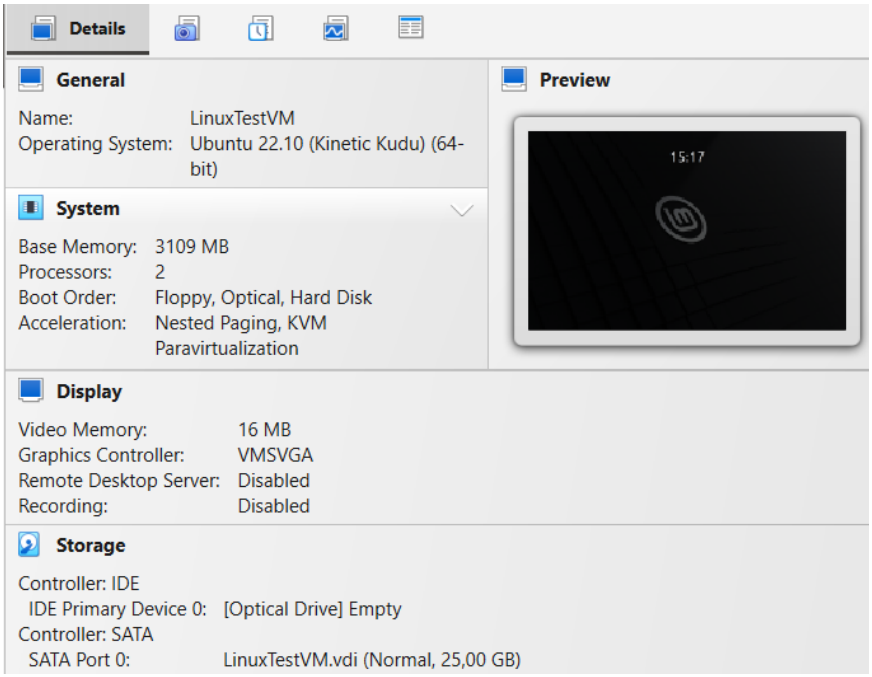
Figure 3: Task Manager showing virtualization



Figure 2: VirtualBox installation completed

STEP 2: Creating the Linux Virtual Machine

Inside VirtualBox, we created a new virtual machine named **LinuxTestVM**. We selected Linux Mint (64-bit) as the operating system type. We configured the Virtual Machine with **3109 MB RAM** and a **24 GB virtual hard disk drive**. And equipped with 2 processor cores.



Explanation:

The Virtual Machine determines how much memory and storage the virtual machine can use.

Figure 5: VirtualBox VM configuration showing RAM, CPU, and storage settings for LinuxTestVM.

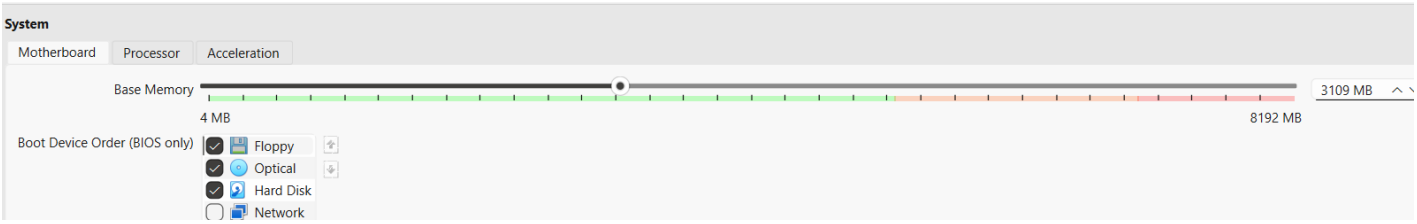


Figure 6: System settings displaying 3109 MB RAM allocation

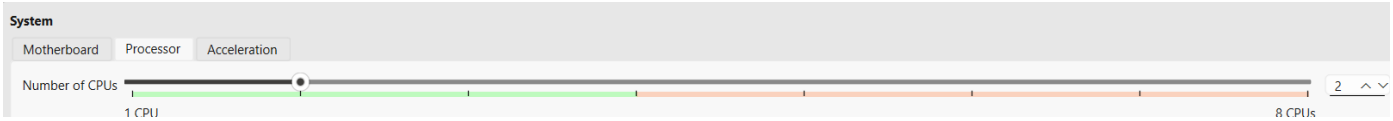


Figure 7: System settings displaying 2 virtual CPUs

STEP 3: Installing the Linux Operating System

We booted the Virtual Machine using the provided Linux ISO. The Linux Mint installer opened, and we followed the guided installation steps: language selection, keyboard layout, installation type, and user setup, after installation, the virtual machine rebooted into the Linux Mint desktop.



Figure 8: Linux Mint boot screen displayed after launching the virtual machine.

Explanation:

This step installs the actual operating system inside the Virtual Machine, allowing us to interact with Linux normally.

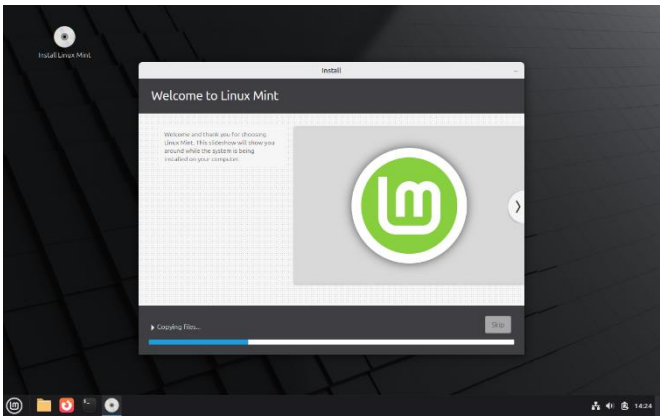


Figure 10:Linux Mint installer welcome window confirming the start of the installation process.

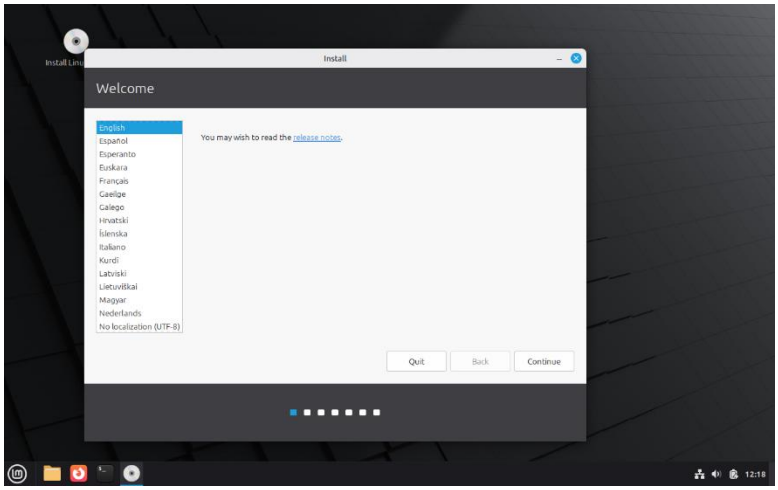


Figure 12: Language selection screen during the Linux Mint installation.

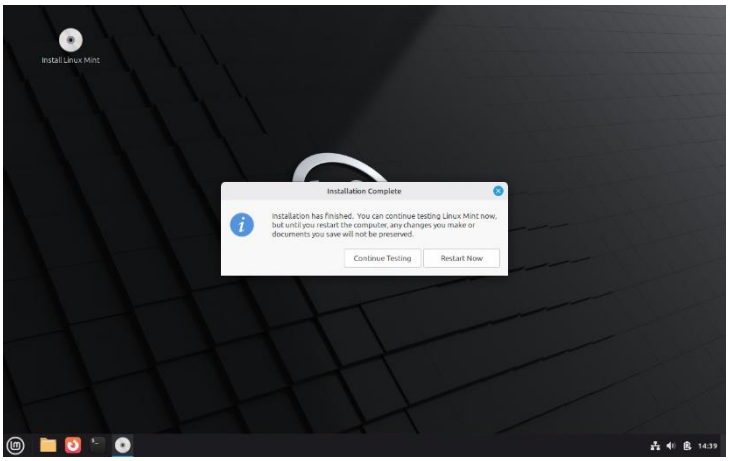


Figure 9: Confirmation that the installation process has completed.

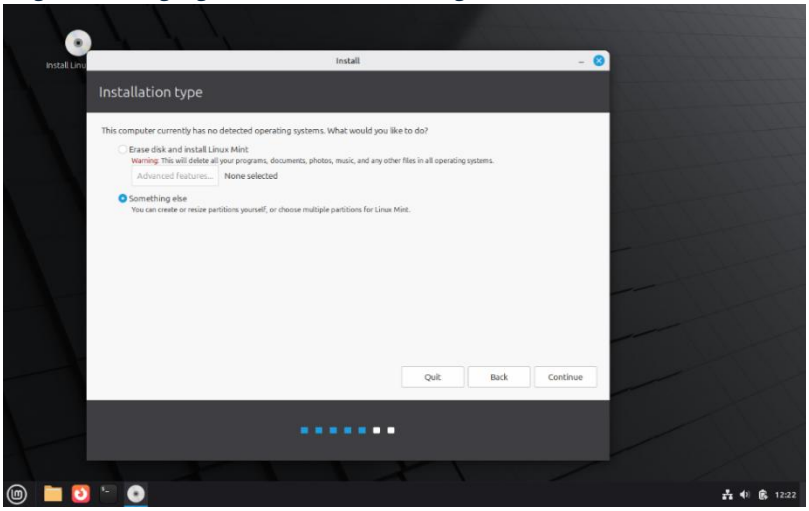


Figure 11:Installation type selection screen for configuring disk partitioning.

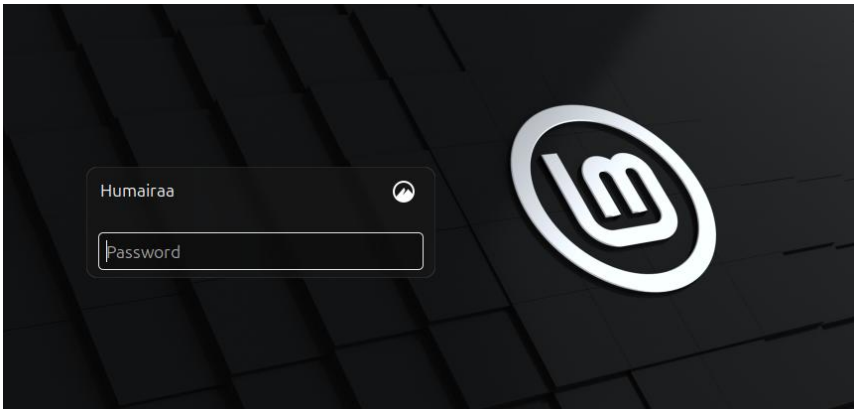


Figure 13:Login screen after installation, showing the user account created (Humairaa).

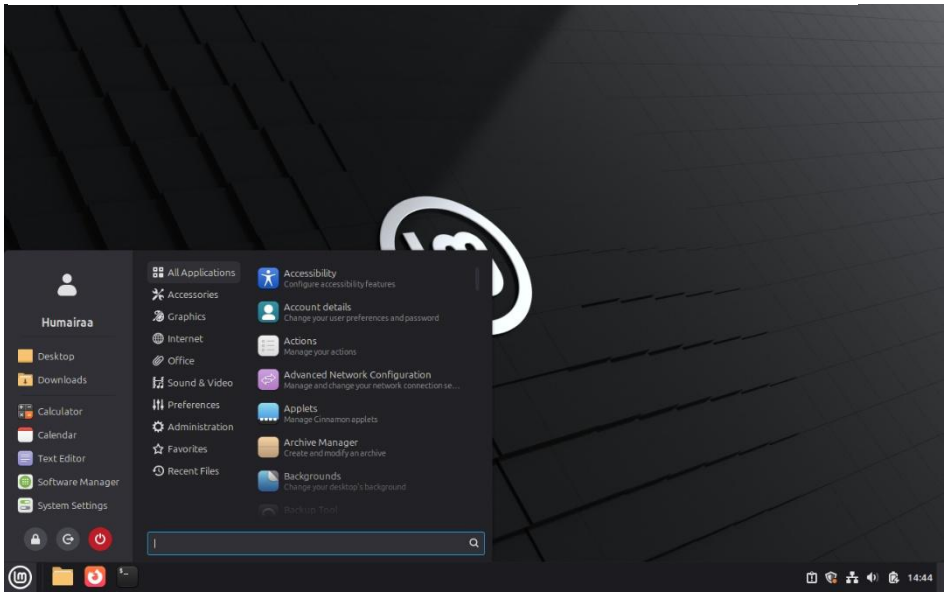


Figure 14: Linux Mint desktop environment after successful installation, with the application menu open.

STEP 4: Displaying System Information

Inside the Linux Mint Virtual Machine, we opened the Terminal and ran the required system-information commands. These commands verify that the Virtual Machine is correctly configured and running the expected Linux environment.

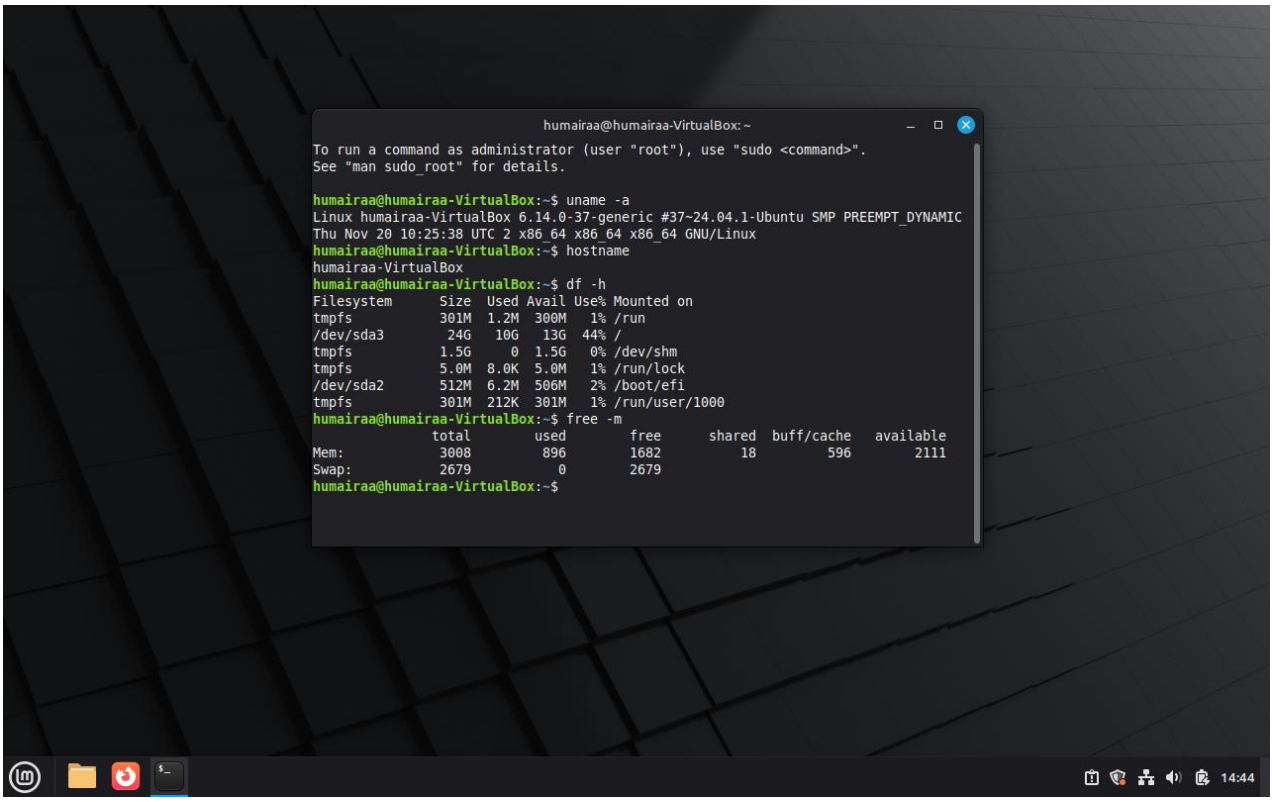


Figure 15: Terminal output showing Linux kernel information using the uname -a, hostname, df -h and free -m command.

Explanation:

- **uname -a** — Shows the Linux kernel version, architecture, and system details. *This confirms the VM is running a genuine 64-bit Linux kernel.*
- **hostname** — Displays the machine's network name. *This proves the commands were run inside the VM (humairaa-VirtualBox), not the host machine.*
- **df -h** — Shows disk usage in human-readable format. *This verifies the 24 GB virtual hard disk was created and is being used.*
- **free -m** — Shows RAM usage in megabytes. *This confirms the VM is using the allocated 3080 MB (3 GB) RAM.*

STEP 5: Cloud Storage Upload and Download

Inside the Linux Virtual Machine, we opened the Firefox browser and logged into OneDrive. We uploaded a test file from the host machine called **LinuxText.txt**. After confirming the upload, we downloaded the same file inside the Virtual Machine and opened it successfully.

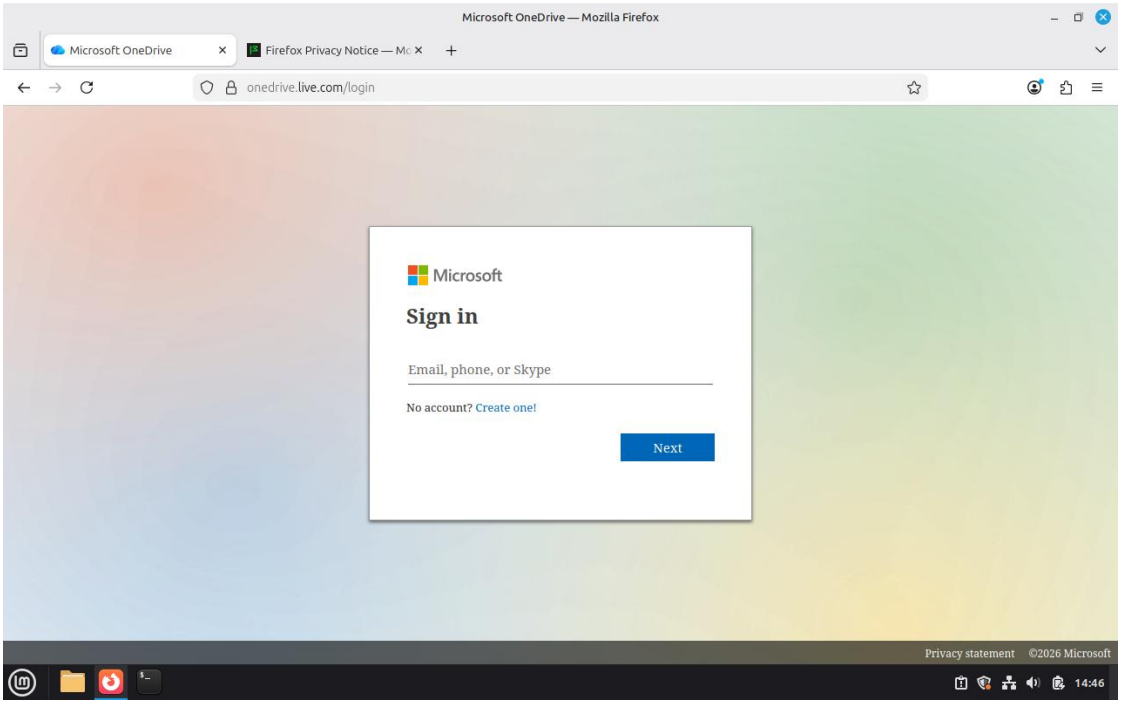


Figure 16:OneDrive login page accessed through Firefox inside the Linux virtual machine.

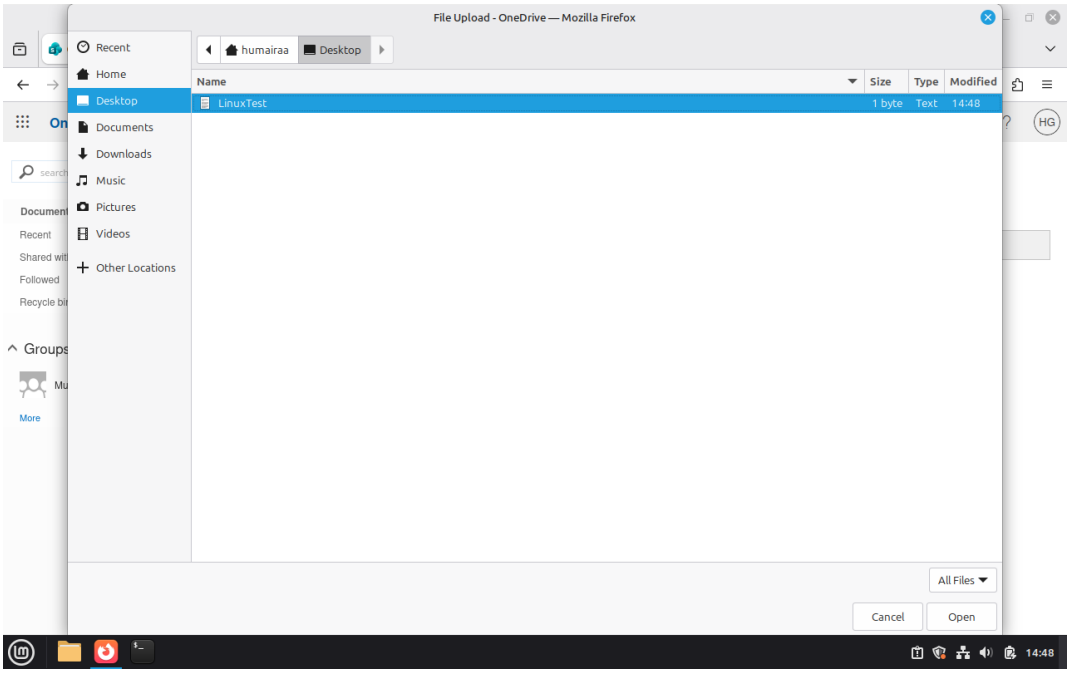


Figure 17:Uploaded file (LinuxText.txt) visible in OneDrive, confirming successful cloud upload.

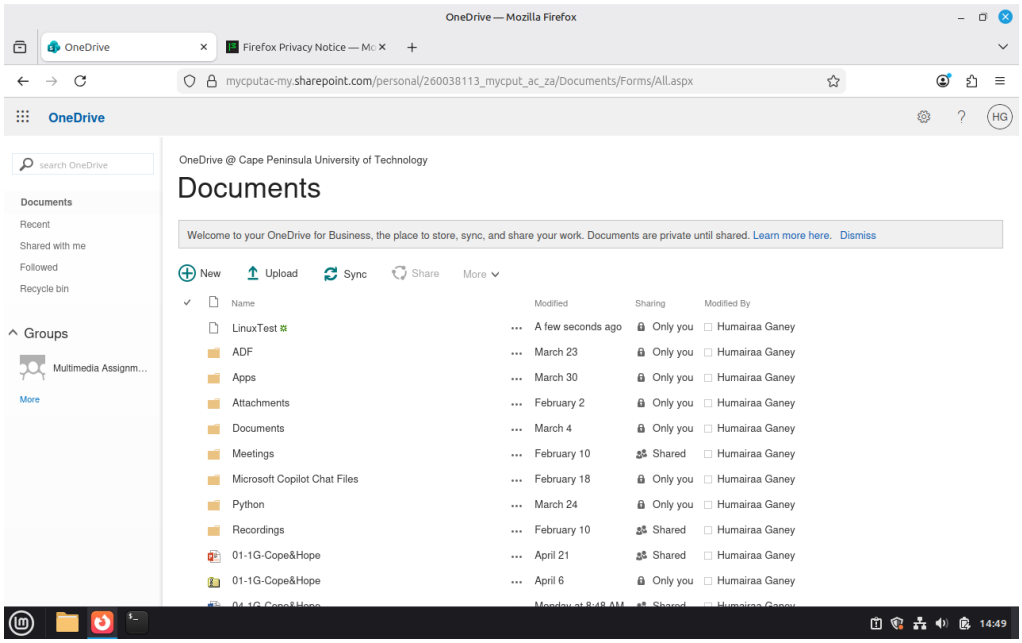


Figure 18:OneDrive folder structure accessed through Firefox inside the Linux virtual machine.

Explanation:

This demonstrated cloud storage functionality in the Virtual Machine, uploading from the one environment and accessing the same file from another.

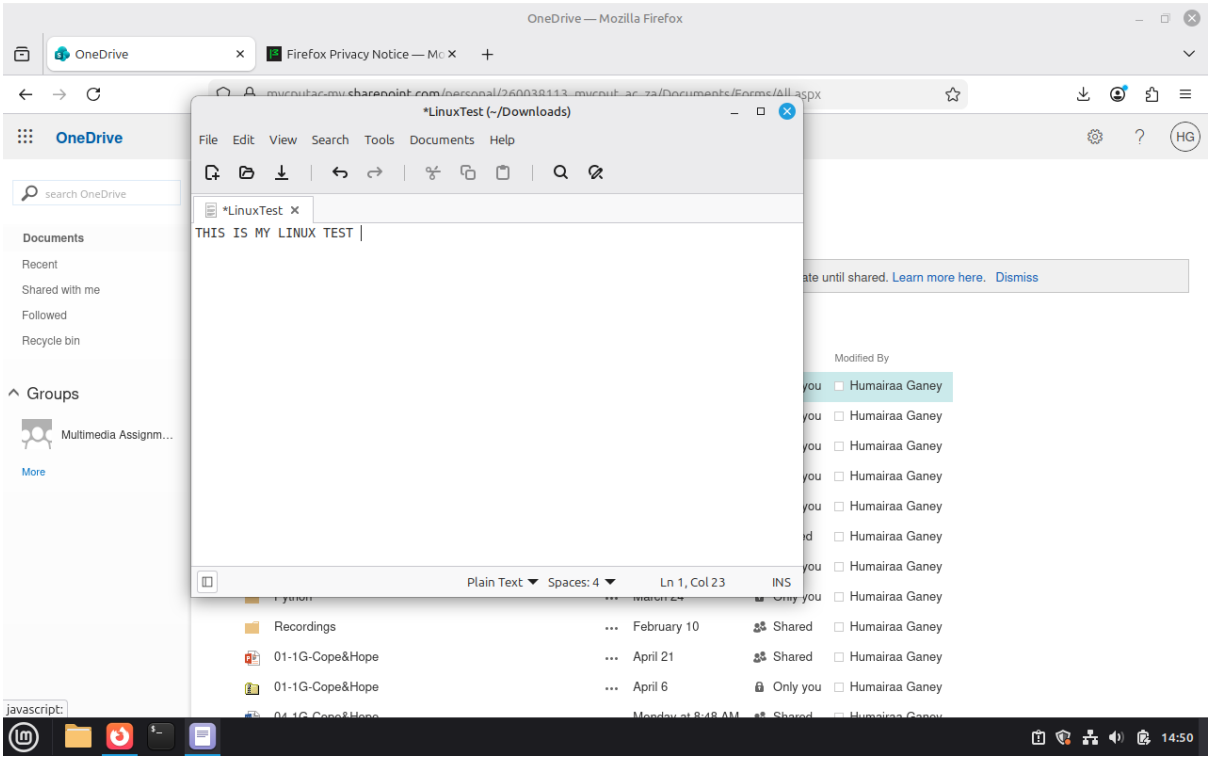


Figure 19:Downloaded cloud file opened inside the Linux VM, confirming successful cross-environment access.

Group Reflection

Virtualization provides a safe, flexible environment for learning without needing multiple physical machines. It allows students to experiment, break things, and reset easily without harming the host operating system. Cloud storage adds convenience by enabling file access from any device and improving collaboration in group work. Together, virtualization and cloud computing make computer labs more efficient, reduce hardware costs, and support learning practices.